Ankitha Betala

Propulsion CFD Engineer
Aerospace simulation Engineer

Email : betalaankitha@gmail.com Ankitha Betala
Phone : +91-6300313493

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - 2026	M.Tech (Aero Thermal)	Indian Institute of Technology, Kanpur	6.5/10
2023	B. E (Aeronautical Engg)	Hindusthan Institute of Technology, coimbatore	8.86/10
2019	Board of Intermediate Education(XII)	Sri chaitanya Jr.Mahila Kalasala,Vijayawada	9.92/10
2017	Board of Secondary Education(X)	Andhra pradesh Residential School,Musunuru	9.5/10

Technical Skills

- **Programming:** Python, MATLAB (simulink), C, LaTeX
- **CFD & Propulsion:** ANSYS Fluent, Internal Ballistics, Solid Rocket Motors, NASA CEA, OpenMotor, BurnSim, Compressible Flow, External Aerodynamics, Conjugate Heat Transfer, UDF Development
- **FEA:** ANSYS Mechanical, Static Structural Analysis, Thermal Analysis, Fluid-Structure Interaction (FSI)
- **HPC & Computing:** High Performance Computing (HPC), Linux Environment, Remote Cluster Computing, MobaXterm
- **Tools:** ANSYS Workbench, NI LabVIEW

Professional Experience

Engineer II (*Jan'26- Present*)
Offered by: INVENTGRID INDIA PRIVATE LIMITED (IG Defence)
Office: First Floor, D-23, Sector 63, Noida, Delhi-NCR 201309 Department: New Development – CFD Analysis

JWALA Defence Project

- Performed propulsion nozzle CFD using ANSYS Fluent, analysing flow characteristics and plume–external-flow interactions.
- Applied the RANS SST k-ω turbulence model to analyse pressure and velocity distributions.
- Investigated shock-wave behaviour and exhaust plume development.
- Post-processed CFD results to evaluate flow fields and plume interactions.
- Conducted conjugate heat transfer (CHT) analysis.
- Prepared technical reports and engineering documentation.

KAL RATO Booster Project

- Post-processed KAL RATO test data to extract key performance parameters and assess booster performance for a 250 kg payload application.
- Calculated required thrust and total impulse for rail inclinations of 15°–20° and target release speeds of 160–190 km/h.
- Performed CFD analysis of rocket motor chamber and nozzle flows using ANSYS Fluent, investigating flow characteristics and plume interactions.
- Conducted conjugate heat transfer (CHT) analysis to investigate coupled fluid-flow and thermal behaviour and evaluate temperature distribution and heat transfer in the motor/nozzle system.

Research Interests

- High-Speed Compressible Flows, Shock-Wave Physics, and Propulsion CFD
- Gas Turbine Aerodynamics, Thermal Management, and Heat Transfer
- Conjugate Heat Transfer (CHT) and Nozzle Thermal Analysis
- Hydrogen Combustion, Combustion Instabilities, Thermoacoustics
- Reacting-Flow CFD, Combustion Dynamics, Acoustic–Flame Interaction
- Combustion Stability, High-Fidelity Simulations, Data-Driven Modeling

Key Projects

M.Tech Thesis: Conjugate Heat Transfer Analysis of a Circular Nozzle with Passive Cooling (*Aug'2024 - 2025*)
Proposed by: Aeronautical Development Establishment (ADE), DRDO
Supervisor: Dr. Abhijith Kushari — Indian Institute of Technology Kanpur

- Conducted numerical investigation of conjugate heat transfer in a circular nozzle with passive cooling using **ANSYS Fluent**.
- Studied the coupled effects of reacting flow, heat conduction, and convection to evaluate nozzle thermal performance.
- Analyzed temperature distribution, heat flux, cooling effectiveness, and thermal gradients within the nozzle structure.
- Performed simulations on High Performance Computing (HPC) clusters using Linux-based remote computing environments.

Welding of Dissimilar Aluminium Alloys (Al7075-Al6063) (*Jan'23 – May'23*)
Queen India Engineering Services, Chennai

- Investigated TIG welding of dissimilar aluminium alloys using ER4043 filler material.
- Evaluated mechanical properties through hardness, impact, and microstructural analysis.
- Performed non-destructive testing (NDT) to identify surface and internal defects.

Course Projects

- **Temperature Measurement of Wind Driven pool fire using Labview** (*April 15,2024*)
  Guide: Dr.Kamal Poddar | Department of Aerospace Engineering, Indian Institute of Technology,Kanpur.
  - Developed a LabVIEW interface for temperature acquisition and monitoring in pool fire experiments.
  - Performed data acquisition, processing, and temperature measurement analysis.
- **Transient Thermal Analysis Of a Sphere With an Enclosure** (*Jan'2024-May'2024*)
  Guide:,DR. MOHAMMAD IBRAHIM SUGARNO | Department of Aerospace Engineering, IIT Kanpur.
  - Conducted transient thermal analysis using ANSYS Workbench.
  - Investigated temperature evolution and heat transfer characteristics of a solid sphere enclosed by an air domain.
- **Finite Element Analysis Convergence and Mesh Independence** | Certificate

Internships

Solid Rocket Motor Thrust Vector Control (*Oct'22- 2023*)
Offered by: Defence Research and Development Organisation(DRDO)
Guide: SV Raja Goutham,Scientist 'E',Advanced Systems Lab, DRDO ,Ministry of Defence, Govt.of India ,Hyderabad-500058

- Acquired Theoritical and Analytical Knowledge in Rocket Propulsion
- Understand solid rocket motor dynamics and thrust vectoring principles.
- Enhance rocket trajectory control and stability.
- During the internship programme,I gained knowledge about how pressure varies at different points of the nozzle.
- designed elliptical convergent divergent nozzle in autocad and subdivided into three parts.
- studied the methods of thrusting vectoring and how they are operated

Relevant Courses

Heat Transfer	Introduction to Virtual Instrumentation
Fundamentals of combustion	Measurements and Data Analysis
Combustion Diagnostics	Composite material and Structures
Problem solving and Python Programming	Numerical Methods
Finite Element Methods	Computational Fluid Dynamics
Convective heat transfer	Rockets and Missiles
Principles of Management	Experimental Stress Analysis

Positions of Responsibility

- **Teaching Assistant,Engineering Graphics (Indian Institute of Technology, Kanpur)** (*Aug'2024-Jul2025*)
  - Worked as a Teaching Assistant in Engineering Graphics, compulsory for the junior undergraduates.
  - Responsible for the design of written and computer-based assignments for the course.
  - Mentored 30 UG students and reviewed their continuous progress
- **Volunteered in ACGT Conference** (*Aug'2024*)
  - Guided **80** participants in acclimatizing to the Environment of the Institute.

Scholastic Achievements

- Secured **All India Rank 1017** in **Gate 2023.**
- Received the **Prathibha puraskar award in 2017 on SSC merit, by Government of Andhra pradesh.** in 2017.
- Received the **Ugam legrand scholarship on Intermediate merit basis,2019-2023.**